Nonlinear Functions, Function Notation & Equivalent Expressions

ClipSAT Printable Practice Suite (clipsat.org) • Topic 8.01 • 20 SPRs + 25 MCQs • Difficulty Progression • Official Digital SAT Curriculum

Part I — Student-Produced Response Questions (SPRs): Foundational Fluency (1–7)

1 [Foundational] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

If $f(x) = 2x^2 - 3$, what is $f(-2)$?

2 [Foundational] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

If $f(x) = 5|x - 1| + 6$, what is $f(7)$?

3 [Foundational] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

A function u satisfies the identity $u(s - 2) = 13s + 14$ for all real s. What is the value $u(10)$?

4 [Foundational] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

If $f(3) = -11$, and $g(x) = f(x) + 1$, what is $g(3)$?

5 [Foundational] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

If $f(x) = 6|x - 1| + 9$, what is $f(-7)$?

6 [Foundational] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

If $f(x) = 4|x + 8| - 4$, what is $f(-5)$?

7 [Foundational] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

Function g is defined by $g(x) = x^2 + 1$. What is $g(g(2))$?

Part I — Student-Produced Response Questions (SPRs): Intermediate Application (8–14)

8 [Intermediate] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

If $f(x) = 3x^2 - 2x + 1$, what is $f(-1)$?

9 [Intermediate] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

Consider the quadratic function $u(s) = 10s^2 - 15$ over the closed interval $[8, 11]$. What is the average rate of change $\Delta u/\Delta s$ on $[8, 11]$?

10 [Intermediate] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

If $f(3) = -17$, and $g(x) = f(x) + 4$, what is $g(3)$?

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11 [Intermediate] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

Let the functions F and G be defined by $F(w) = 15w - 12$ and $G(w) = w^2 + 2$. What is the composite value $F(G(18))$?

12 [Intermediate] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

If $f(x) = 2|x + 9| - 1$, what is $f(10)$?

13 [Intermediate] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

If $f(x) = -x^2 - 4$ and $g(x) = 5x + 6$, what is $f(g(1))$?

14 [Intermediate] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

What is the smallest value of x for which $f(x) = \sqrt{x + 1}$ is defined?

Part I — Student-Produced Response Questions (SPRs): Challenging Synthesis (15–20)

15 [Challenging] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

What is the minimum value of $f(x) = 5|x - 6| + 4$?

16 [Challenging] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

If $f(x) = 1|x + 7| - 8$, what is $f(-10)$?

17 [Challenging] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

The function h is defined by $h(x) = 2x + k$. If $h(3) = 11$, what is k?

18 [Challenging] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

What is the smallest value of x for which $f(x) = \sqrt{x - 4}$ is defined?

19 [Challenging] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

If $f(x) = x^2 - 8$ and $g(x) = -2x + 8$, what is $f(g(2))$?

20 [Challenging] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

If $f(-3) = 14$, and $g(x) = f(x - 3)$, what is $g(0)$?

Part II — Multiple-Choice Questions (MCQs): Foundational Fluency (21–28)

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21 [Foundational] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

What is the minimum value of $f(x) = 1|x - 9|$?

a. -3

b. 0

c. -6

d. 6

22 [Foundational] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

What is the minimum value of $f(x) = 2|x + 9| - 7$?

a. 0

b. -9

c. -7

d. -8

23 [Foundational] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

For what value of x is the function $f(x) = \frac{1}{x-5}$ undefined?

a. 5

b. 6

c. 1

d. -3

24 [Foundational] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

For what value of x is the function $f(x) = \frac{1}{x-9}$ undefined?

a. 18

b. 3

c. 8

d. 9

25 [Foundational] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

If $f(8) = 11$, and $g(x) = f(x) + 5$, what is $g(8)$?

a. 8

b. 22

c. 16

d. 10

26 [Foundational] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

What is the minimum value of $f(x) = 1|x - 7| - 1$?

a. 0

b. 6

c. -10

d. -1

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27 [Foundational] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

What is the minimum value of $f(x) = 8|x + 5| + 5$?

a. 2

b. 5

c. 12

d. -1

28 [Foundational] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

If $f(x) = 5|x + 1| + 3$, what is $f(-8)$?

a. 41

b. 39

c. 35

d. 38

Part II — Multiple-Choice Questions (MCQs): Intermediate Application (29–37)

29 [Intermediate] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

What is the minimum value of $f(x) = 7|x + 5| - 10$?

a. -16

b. -8

c. -5

d. -10

30 [Intermediate] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

What is the minimum value of $f(x) = 6|x + 7| + 3$?

a. -1

b. 11

c. 3

d. -4

31 [Intermediate] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

If $f(7) = 17$, and $g(x) = f(x) - 1$, what is $g(7)$?

a. 24

b. 19

c. 16

d. 18

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32 [Intermediate] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

For what value of x does the function $f(x) = 2x + 8$ equal 0?

a. -4

b. -2

c. 2

d. 4

33 [Intermediate] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

If $g(x) = \sqrt{x + 9}$, what is the smallest real number in the domain of g?

a. -9

b. -3

c. 0

d. 3

34 [Intermediate] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

If $f(x) = 5x - 3$ and $f(a) = 22$, what is the value of a?

a. 4

b. 5

c. 6

d. 7

35 [Intermediate] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

If $k(x) = \frac{12}{x}$, what is $k(k(4))$?

a. 3

b. 4

c. 6

d. 12

36 [Intermediate] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

Which expression is equivalent to $(x + 7)(x - 7)$?

a. $x^2 - 14x + 49$

b. $x^2 - 49$

c. $x^2 + 49$

d. $x^2 + 14x - 49$

37 [Intermediate] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

If $f(x) = 5x^2 - 5$ and $g(x) = -2x + 2$, what is $f(g(3))$?

a. 71

b. 66

c. 74

d. 75

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Part II — Multiple-Choice Questions (MCQs): Challenging Synthesis (38–45)

38 [Challenging] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

If $f(-7) = -18$, and $g(x) = f(x) - 3$, what is $g(-7)$?

a. -21

b. -12

c. -25

d. -16

39 [Challenging] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

What is the minimum value of $f(x) = 1|x - 7| - 8$?

a. 1

b. -10

c. -1

d. -8

40 [Challenging] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

For what value of x is the function $f(x) = \frac{1}{x}$ undefined?

a. 0

b. 9

c. 3

d. -1

41 [Challenging] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

If $f(x) = 4|x - 5| - 8$, what is $f(4)$?

a. -12

b. 2

c. -4

d. -13

42 [Challenging] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

If $f(x) = 3x - 5$, what is the value of $f(4)$?

a. 5

b. 7

c. 9

d. 12

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43 [Challenging] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

Which of the following values of x is NOT in the domain of $h(x) = \frac{5}{x-7}$?

a. -7

b. 0

c. 5

d. 7

44 [Challenging] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

A function f satisfies $f(3) = 11$ and $f(5) = 17$. If f is linear, what is $f(1)$?

a. 3

b. 5

c. 7

d. 8

45 [Challenging] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

Which expression is equivalent to $(x + 3)(x - 3)$?

a. $x^2 + 6x - 9$

b. $x^2 + 9$

c. $x^2 - 6x + 9$

d. $x^2 - 9$

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